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RESEARCH ARTICLE

Deep Learning-Based Real-Time Driver Cognitive Distraction Detection

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The study was consistent with the principles from the Declaration of Helsinki and the ethical laws that were applied in France at the time of the experiment.

ABSTRACT Driver distraction is one of the main causes of traffic accidents. While there are different types of distraction (manual, visual, cognitive), cognitive distraction is particularly challenging, being only partially related to visual features detectable through cameras or an eye tracker system. Moreover, since cognitive distraction is not a point in time phenomenon, spotting this kind of distraction requires the processing of a certain time interval, which poses a further challenge for real-time performance. After a data collection campaign with $N = 42$ subjects undertaking a twenty-question task (TQT) in a driving simulator, we developed a driver cognitive distraction detection system, with the goal of filling in some key gaps we identified in the literature towards real world deployment. First, we assessed the effectiveness of state-of-the-art time series-oriented deep learning models in learning features from 60 Hz raw input signals, thus implementing an end-to-end machine learning approach, without manual feature engineering. We demonstrated that such models are able to classify time-windows as small as 0.5 seconds, and are also more robust to sensor failures. Second, also with the support of AI explainability, we showed that processing vehicular data is fundamental to ensure performance, while physiological signals provide a less important, but still useful, contribution. Third, through a between- and within-subject design comparison, we showed that eye-tracker and, particularly, physiological signals are much more prone to inter-individual variability, thus overfitting. This is fundamental to consider for commercial deployment, as it would require fine-tuning the system with data from the actual end-user. Fourth, we quantitatively measured the effect of such variability for all types of signals, demonstrating its huge relevance, and shown that deep learning models dedicated to time series processing are better able to generalize across users than the more commonly employed shallow machine learning models. Finally, with a focus on in-vehicle deployability, which is of significant industrial interest, we measured also such metrics as model size, inference time, and energy consumption, showing feasibility on two embedded platforms, which is a key advancement towards on-board deployment of robust, real-time cognitive distraction detection systems.

INDEX TERMS Deep learning, driver cognitive distraction detection, embedded deployment, eye-tracker, physiological signals, SHAP analysis, timeseries processing, vehicular signals.

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I. INTRODUCTION

Issues with the driver's cognitive state (e.g., inattention, distraction, drowsiness) are a major cause of road accidents [1]

and it has been estimated that can be responsible for more than 35% of road fatalities [2]. In the United States, the National Highway Traffic Safety Administration (NHTSA) reported that 3,142 distracted driving fatalities occurred in 2020 [3].

While driver's distraction is generally deemed an important safety concern [4], its related concepts are frequently used in different terms in literature (e.g., [5], [6]). In this work, we follow the definition proposed by Regan et al. [7], which joins accuracy and generality: "Driver distraction is the diversion of attention away from activities critical for safe driving toward a competing activity".

While literature considers different forms of distraction, three are the classes usually employed in detection algorithms (e.g., [1], [8]): manual distraction (in which the driver takes his/her hands off the wheel, e.g., for drinking, eating); visual distraction (in which the driver looks away from the road, e.g., for reading, watching the phone); cognitive distraction (in which the driver's mind is not fully focused on the driving task, e.g., for talking, thinking).

Manual and visual distractions have been extensively studied in the literature, also thanks to the use of computer vision and/or machine learning (ML) systems able to effectively process signals from camera and eye-tracking sensors (e.g., [9], [10]). Cognitive distraction (intuitively, "mind off-road"), on the other hand, is less correlated with visual features (it does not necessarily take a driver's eyes off the road, as it concerns "looking but not seeing"), thus is harder to detect [11]. Cognitive distraction is prevalent and particularly challenging to detect [12], [13]. Its complexity arises from different attentional priorities [14], which can lead to stimuli being incorrectly recognize or inadequately responded to [15]. Ground truth is also difficult to achieve, as cognitive distraction is usually labeled as such during the execution of a cognitive distraction task, which does not necessarily imply that a subject is actually distracted (i.e., attention is diverted from the task of driving). Conversely, a driver may become cognitively distracted even without engaging in a specific external task. This introduces uncertainty in the data to be analyzed. Addressing cognitive distraction typically requires also the use of physiological sensors that can be integrated into a vehicle, such as those measuring the driver's electrodermal activity and/or heart rate.

To effectively study cognitive distraction, it is crucial that collected data be processed over sufficiently long-time windows to identify meaningful patterns (e.g., [16]). This poses challenges in terms of real-time performance, which is an unavoidable requirement for in-vehicle system deployment [17].

We have faced this research context in the Hi-Drive collaborative project [18], in which we have developed a Driver Distraction Detection (DDD) system focused on cognitive distraction. The system has been developed in a simulation environment, given the risks associated with real-world data collection for this task. The main goal of Hi-Drive is to advance automated driving (AD) systems "Conditional

Automation" towards "High Automation", demonstrating in large-scale trials the reliability of AD functions (ADFs). This is targeted by extending the operational design domain (ODD) and thus reducing the frequency of takeover requests (TORs) [19], implementing technology enablers to improve state-of-the-art ADFs. DDD systems are even more relevant at higher automation levels, because of the need to elaborate effective TOR issuing strategies [1], [20].

TABLE 1. Main gaps addressed.

Gap	Item	Gjoreski et al. [16]	Misra et al. [11]	Ours
1	Shallow vs deep ML	✓	Shallow only	✓
	Time-window size	20 – 80 s	1 s	0.5 - 20 s
	Signal types	Camera, eye-tracker, physiological	Vehicular, eye-tracker, physiological	Vehicular, eye-tracker, physiological
2	Train/test subject split	✓	-	✓
3	Deployability metrics	-	-	✓

The cognitive distraction detection topic is wide, with significant implications in terms of safety and privacy. The novelty of this paper consists in addressing three main gaps emerging from the literature (e.g., Gjoreski et al. [16] and Misra et al. [11] – please see Table 1 for a synthetic synopsis). First, most of the machine learning (ML) models developed for cognitive distraction are shallow (e.g., decision trees, random forests, support vector machines), as they process a limited number of features extracted from the sensor signals based on expert knowledge. On the other hand, we are interested in exploring the performance of state-of-the-art deep learning (DL) models specifically designed for time series processing. These models learn the best features directly from the input raw signals during the training process, without resorting to feature engineering. For such models, we investigate the usefulness of different input data types (vehicular signals, eye-tracker, physiological data), also resorting to explainability techniques [21]. Only seldom and marginally does the literature employ vehicular signals. But vehicular data are easily available in vehicles, while eye-tracker data require a costly set-up, and physiological data are very invasive for the user. We are also interested in verifying if these DL models can detect distraction patterns in very short time windows, minimizing latency. As second, major gap, we intend to assess the generalizability of the models, testing their performance on a set of drivers different from the one used for training. Gjoreski et al. [16] performed this separation, but on large time windows (20-80 s), without vehicular signals and without measuring the difference with respect

to un-separated testing. Using test samples from subjects also employed for training would imply that the deployed system is trained with samples from the end user. But this approach risks being unrealistic from an industrial perspective and could lead to an over-optimistic assessment due to overfitting to the dataset subjects. Thirdly and finally, again especially considering an industrial perspective, we intend to explore the deployability of the models in their target environment, thus measuring embedded computing metrics (e.g., latency, model size, energy consumption) on low-cost microcontrollers deployable in a vehicular architecture.

We believe that these research questions, while not exhaustive, are fundamental and urgent to address, and may have a major impact on future research as ever more powerful ADFs are continuously being developed. We have thus set up a relatively simple, static simulation environment that, while lacking vehicle movement and featuring a single distraction task and driving scenario, made it possible to minimize the experimental factors and ensure a timely and cost-effective analysis, that could have a significant impact on future research.

The remainder of the article is organized as follows. Section II discusses the related work. Section III introduces the conducted experiments, presenting user test settings and procedures, signal collection and post-processing, and dataset preparation and split. Section IV briefly presents the deep learning models and their tuning. Section V discusses the experimental results, while Section VI draws conclusions and suggests possible directions for future research, especially in light of the new findings.

II. RELATED WORK

The Introduction has already mentioned the three main types of distraction and the lack of a common agreement on their definition and characterization. As general overviews of the field, we highlight Kashevnik et al. [1], that provide an extensive literature review and framework of DDD methods, and Santosh Kumar and Jayanthi Kannan, [22], that address computer vision techniques for driver monitoring.

Within this broad area, our focus is on cognitive distraction. In literature, many secondary tasks have been used to induce cognitive distraction, such as conversation (e.g., Crundall et al. [23]), speech comprehension tasks (e.g., Kubose et al. [24]), and mathematical tasks (e.g., Chaparro et al. [25]). Misra et al. [11], which represents a key reference in the area, implemented a spoken task with a series of statements played through the speakers after a constant period of five seconds, to which the driver responded by identifying and telling subject and object of the question and answering to it. In our study, the distraction source is a twenty-question task (TQT), in which the subject has to guess an object by asking a maximum of 20 questions, each one responded by the supervisor simply in the form of yes/no Horrey et al. [26]. TQT is widely used in cognitive psychology and neurophysiology, as well as in driver's cognitive distraction experiments Merat et al. [27].

DDD systems typically employ three main types of input data: driving data, physiological data, and eye-tracker data. The first category concerns vehicle kinematics, which are the motion characteristics of the vehicle. These parameters are sensitive to driver distraction, which may cause changes in acceleration, braking, gear changing, and steering patterns (e.g., Castignani et al. [28]). Additional indicators are given by performance defects in maintaining speed and lane keeping (e.g., Aksjonov et al. [29]). Other useful information on driver distraction comes from investigating physiological responses to cognitive load, which reflects the extent to which a driver is cognitively distracted from driving [30]. Several signals in this area can serve as indicators of cognitive load and cognitive distraction. The electroencephalogram (EEG) measures brain electrical activity (e.g., Li et al. [31]); the electrocardiogram (ECG) measures variations in heart rate (e.g., Deshmukh and Dehzangi [32]) (heart rate can be estimated also via video processing, e.g., [33], [34]); electromyogram (EMG) traces changes in muscle tension (e.g., Kumar et al. [35]); electrodermal activity (EDA) measures changes in skin conductance (e.g., Healey and Picard [36]). Eye-tracker systems are another significant tool for driver distraction or fatigue detection, by tracing eye closure, pupil movement, eyeblink, and gaze position (e.g., Rehman et al. [37], Liu et al. [38]). Later studies have demonstrated that combining different types of signals improve accuracy. For instance, Gjoreski et al. [16] use eye-tracker data, physiological data, and emotional responses detected through cameras to identify distractions in time windows of different sizes. Misra et al. [11] use physiological, eye-tracker, and driving signals and quantitatively measure the contribution of each different data type and their combinations.

A key factor to consider for actual vehicular deployment is that DDD systems are characterized by different levels of intrusiveness to the user comfort and privacy, due to the collection, processing, and possible storage and transmission of sensitive information about the driver (e.g., Kashevnik et al. [1]). Zhou et al. [39] have proposed a privacy-preserving federated-learning approach, allowing to build a ML model without sharing sensitive data at centralized level. A recent study by Chu et al. [40] aimed at gathering insights into Chinese drivers' attitudes towards DMS privacy issues and their willingness to adopt this technology. The authors observed that Chinese drivers, in general, are less worried about privacy and data insecurity, hold a favorable view of driver monitoring systems, and express a degree of willingness to use them. However, they conclude that further study is necessary to ascertain readiness to use in real-world scenarios.

Different ML solutions have been studied for distraction detection. We can broadly divide them into shallow ML techniques, which are more suited for processing data points (e.g., signals at specific timestamps or features extracted through domain-specific methods), and DL techniques, which typically provide the state-of-the-art solution for end-to-end ML processing of sequential/grid input.

Goel et al. [41] extract features from a wrist-worn wearable device and process them with various shallow ML models, such as decision tree (DT), Naïve Bayes (NB), Support Vector Machines (SVM), and Random Forests (RF) to detect manual distraction, highlighting the superiority of RF for the task. Similarly, Lee et al. [42] investigated hand movements captured by acceleration sensors in a smartwatch, which were then input into an SVM classifier. Going to cognitive distraction, Misra et al. [11] compared on a data-point classification task different models like Naïve Bayes, linear SVM, Radial Basis Function (RBF), SVM, K-Nearest Neighbors (K-NN), DT, and RF, showing that RF achieved the best accuracy (91.5%, averaging various road conditions). Gjoreski et al. [16] apply shallow ML techniques to classify time-windowed signals reaching the best performance with Gradient Boost (GB) and eXtreme GB (XGB) (73%).

Deep Learning models are typically applied to image input for detecting visual or manual distraction. The availability of datasets like the State Farm Distracted Driver Detection (e.g., it stores images from 2D dashboard cameras with a driver doing something in the car, such as texting, eating, talking on the phone, makeup, reaching behind) has led to high accuracy in distraction classification, with various types of neural architectures, such as Multilayer Perceptrons (MLP) (97.4%, in [43]), Long-Short Term Memory (LSTM) Networks (99.6%, in [44]). Most of the systems employ Deep Convolutional Neural Networks (CNN). An accuracy of 96.0% is achieved by both Pandey and Muppaleneni [45] and by Ezzouhri et al. [46], which makes a body-part image segmentation as a fundamental pre-processing step. Results above 99% are achieved by Gumaei et al., on a VGG16 model, [47], by Sahoo et al, using a SqueezeNet deployable on a Raspberry Pi 4B device in [48]), and by Sajid et al., using an implementation of EfficientDet, [49].

Differently from visual or manual distraction, cognitive distraction lacks publicly available datasets, which is a significant gap for research. Cognitive distraction sees a clear prevalence of shallow ML solutions. As said, Misra et al. [11] use only shallow ML, while Gjoreski et al. [16] make an extensive comparison between the two types of classifiers, showing that the best shallow models (XGB and GB, with an accuracy of 72% and 73%, respectively) perform better than the best deep ones (eLSTM and spectro-temporal ResNet (STRNet), with 67%). We argue that this comparison should be deepened, in the light of the latest advancements that highlight the CNN's abilities to automatically extract features from signals [50]. More investigation would be needed also on the size of the time window employed to detect distraction, which requires setting a trade-off between accuracy and latency, as large time windows could prevent real-time functioning. Liu et al. [38] assess driver distraction using a 10-second window size, while Nakano and Chakraborty [51] investigate a range from 15 to 70 seconds, Gjoreski et al. [16] analyze performance with different window sizes (from 20 to 80 seconds), while Misra et al. [11] have a different approach,

using a sampling rate of 1 second and include a bunch of secondary features statistically aggregating the values of a signal in that period (e.g., steering standard deviation).

Research has developed also in the direction of detecting drivers' fatigue and drowsiness, primarily analyzing vehicle control signals like steering wheel movement, to avoid the more privacy-invasive cameras. Here, Neural Networks (NNs), Bayes classifiers, and RFs achieve accuracy up to 90% (e.g., [52], [53], [54]). Recent studies have explored physiological signals. For instance, Patel et al. [55], obtained 90% accuracy processing with a Fully Connected Neural Network (FCNN) ECG data recorded under laboratory conditions. Yeo et al. [56], on the other hand, processed EEG data, achieving 99.3% accuracy with an SVM. Hanafi et al. [57] use a Deep Belief Network (DBN) to compute the percentage of eyelid closure over the pupil over time (PERCLOS), which value is compared with a threshold (3 blinks in a 15 second window) to signal sleepiness. Their system is deployable on a mobile app but was not tested in automotive settings.

An important limitation of current literature, especially in view of real-world deployment, concerns system test modalities. State-of-the-art DDD studies usually do not differentiate between training and test users. This assumes that a system is trained using data also from the actual end-user (or that the model's input is largely independent of the user, which is not likely, especially considering physiological signals and different driving styles). This has been recently pointed out by Hasan et al. [58], who propose a unique validation method, named "leaving a set of drivers out" (which extends the well-known "leaving a set of samples out" commonly used in validation), which involves subject-level separation of the samples to prevent overfitting to specific drivers. Gjoreski et al [16] applied test subject separation also for cognitive distraction. However, they considered large time windows (at least 20 s) and did not employ vehicular signals nor measured the difference with respect to un-split subject testing. Moreover, they reported that the most important feature is increased lip movement, which is clearly dependent on the type of cognitive distraction task employed (i.e., answering questions). Building on their experience, we decided to go more in depth into analyzing inter-individual variability when testing a cognitive distraction detection system.

Considering more recent types of neural architectures, Graph Neural Networks (GNNs) have been studied for driver drowsiness detection. Zhuang et al. [59] used a GNN encoder to process frequency features extracted via FFT from an EEG signal. Very recently, Qiao et al., [13], used a stack of Transformer blocks to extract features for cognitive distraction classification exploiting the spatial cross-attention mechanism. The article also proposes (i) a preprocessing technique that effectively integrates temporal information of driver eye movement data and (ii) a fusion model to merge features from the eye movement heat Map and the DashCam Image (DCI). Reported results (96.42% accuracy) are promising, but the study does not involve actual driving operations such as

steering or braking, as driving scenes were simulated solely by letting participants watch DCI videos.

Summarizing, the state-of-the-art overview highlights some gaps worth exploring. These include performance assessment of deep learning models dedicated to timeseries processing, analysis of the types of data to process, and time window size. Moreover, inter-individual variability and generalization should be investigated, since training a system also with data from its final user is not easily feasible for industrial products. Finally, there is a clear lack of studies assessing the deployment of a DDD system in an embedded environment, such as the vehicle, thus keeping into account typical embedded metrics like memory footprint, latency, and power and energy consumption. [46] and [55] address deployability, but with a limited set of metrics (e.g., not measuring power/energy consumption) and not on the cognitive detection task.

III. EXPERIMENT

This section describes the design of the user experiments and the procedure adopted for data collection, also including the driving simulation settings. The study presented in this paper was part of a broader study on level 3 Automated Driving within the Hi-Drive project (in particular, how the Human-Machine Interaction of AD, integrated with the driver's state, can improve a safe transition from automated to manual driving). It is important to precise that this data collection was done preliminarily, thus avoiding any bias from the subsequent ML processing.

A. PARTICIPANTS

Our sample was quite various and balanced, consisting of 42 participants (23 females, 19 males) of age between 22 and 62 years (mean age of 35.64 ± 9.78 years). Participants had on average their driving license for 15.80 ± 10.29 years and drove for about $11,404.76 \pm 8,705.37$ kilometers a year. The study was implemented in accordance with the principles of the Declaration of Helsinki and the ethical laws applied in France at the time of data collection. All participants signed a written informed consent form and received 40 euros for their participation.

The number of 42 users has been chosen based on the results of a statistical power analysis done with G* Power 3.1.9.7 [60]. This choice overcomes the state-of-the-art, as the number of participants in similar works is 9, 24, 26, 40, 41 in [11], [31], [46], [49], and [38], respectively. We argue that the higher number of subjects (68) chosen by Gjoreski et al. [16] is due to the broader extent of their study, which, beside cognitive distraction, also analyzed emotional, sensorimotor and mixed distractions.

B. APPARATUS

The medium-fidelity static driving simulator set up at the VEDECOM labs consists of three screens of 140×230 cm allowing 230° of vision and includes a two-seat cabin, a steering wheel, a set of pedals, three mirrors (i.e., left, rear, and

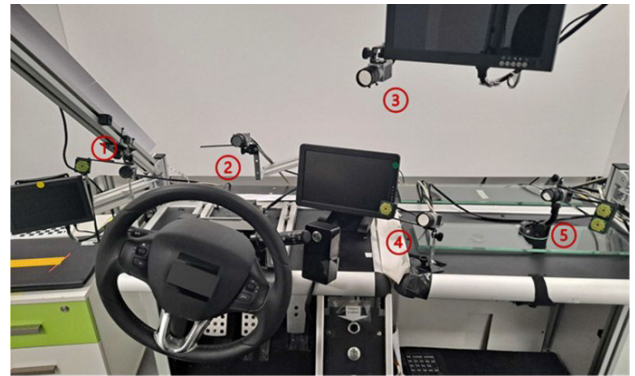


FIGURE 1. In the 5-camera Smarteye Setup, each number indicates the placement of a camera.

right), and a dashboard. The driving simulator ran under SCANer studio 2023.1 (Avsimulation®), which was synchronized with a 5-camera eye-tracking setup (Smart Eye Pro 6.0). Eye-tracker camera positions are shown in Figure 1. The driving simulator and camera signals were sampled at 60 Hz. A BioPac Bionomadix setup with a 3-electrode electrocardiogram (ECG) and Electrodermal activity (EDA) was used to record and sample physiological signals at 2,000 Hz (Biopac MP150 with AcqKnowledge 5).

C. COGNITIVE DISTRACTION TASK

The cognitive load was induced by the Twenty Question Task (TQT) [27], which aims at emulating a conversation with a passenger or a phone call. This task consists of guessing something within a category, in our case an animal or a public figure, by asking a maximum of twenty questions that can only have responses that are “yes” or “no”. This task is known to use cognitive resources such as working memory, problem-solving, and visualization, which is very close to a phone conversation. This task was preferred over the n-back task, even if the n-back task is commonly used in driving studies (for a meta-analysis on the n-back task, see von Janczewski et al. [61]) because the TQT presented more advantages due to its ecological aspect, better mimicking a real-world distractive activity.

D. PROCEDURE

Prior to using the driving simulator, participants received information regarding the goals of the study and completed the informed consent form. The experimenter applied the ECG and EDA electrodes on the participant, verifying that a good signal quality was obtained from the electrode's placement. The ECG electrodes were placed according to ECG placement standards following Einthoven's triangle on the chest with one electrode near V1, one near V5, and earthing below floating ribs, which was found to be the best placement for the QRS complex with a small number of electrodes [62]. The EDA electrodes were placed on the left wrist due to the driving constraints.

At the beginning of the experiment, participants completed a short evaluation of their trust in automated driving by means



FIGURE 2. The driving simulator during the highway drive.

of a slider ranging from 0 (lowest trust) to 100 (highest trust) and a questionnaire of acceptability [63]. At the end of the experiment, participants completed a technophilia questionnaire as well [64].

Participants had to drive on a 2-lane highway with ongoing traffic, being instructed to stay in the right lane (Figure 2). During the study, the experimenter ensured that participants followed instructions to be looking at the road. Additionally, the surrounding environment was intentionally kept quite simple, to minimize any potential distractions.

Participants started with a 10-minute training session, in which they could familiarize themselves with the driving simulator. After the end of the training, participants seamlessly continued driving the car for two consecutive 90-second periods, which constitutes our data sample. After the first period with no cognitive distraction, the TQT cognitive distraction task started.

This design was very important to prevent potential bias from alternating multiple distraction and non-distraction periods. We deemed that the cleanest approach was to start without inducing any cognitive distraction for a not too long period, so to prevent the risk of mind-wandering, boredom, or drowsiness. Also, the second period, during which participants were engaged in a task promoting cognitive distraction, had a duration that prevented the induction of stress, which could have impacted the collected data. The equal time length of the two experimental conditions also allowed the creation of a balanced dataset. Overall, we aimed at minimizing the number of possible factors, thus resulting in a “simple” set-up, so to better control the experiment and allow a cleaner analysis.

E. DATA STRUCTURE

For the sake of the study, we extracted the following data from the driving simulator: Speed (m/s), Steering wheel speed (rad/s), Steering wheel angle (rad), Steering Wheel Torque (Nm), Gas pedal activation (%), Brake pedal activation (N). All these signals were sampled at 60Hz. For the eye-tracking data, real-time and estimated eye-gaze position, head rotation, pupil diameter and the eyelid opening were collected, sampled at 60Hz. For the physiological data, we used the

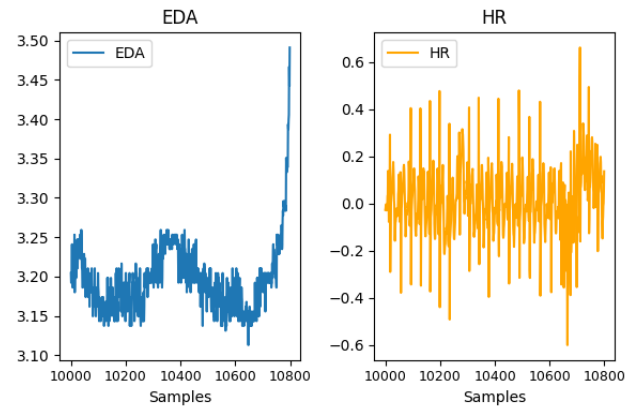


FIGURE 3. Example of EDA and HR signal divergence.

cardiac -rhythm (mV) and EDA response (S), sampled at 2,000Hz.

F. SIGNAL POST-PROCESSING

Raw data from the driving simulator and the ECG were collected with no issues nor need for post-processing. The collection of EDA data was more complicated. A significant number of participants lost the electrodes placed on their wrists due to the movement of the steering wheel and the temperature (about 26-28 °C) in the driving simulator. As an example, Figure 3 shows a divergence in the EDA and heart rate (HR) values towards the end of the series. This can be seen as a limit of our experimental setup. However, it also reflects an objective invasiveness, also in terms of privacy, of physiological data collection. Moreover, it allows to test the distraction detection system in sensor failure conditions as well. Thus, we argue that it represents a realistic challenge towards real-world deployment. In this perspective, we chose to keep the raw input with no signal post-processing. On the other hand, we applied post-processing to the eye-tracking signal because of a significant data loss ($21.28\% \pm 5.56\%$) due to the system, not the user interaction. A linear interpolation was performed on missing data and applied on any missing segment of no more than 150ms, above which data was considered lost. This threshold was chosen according to the literature to be able to interpolate during a complete saccade fixation and saccade process [65]. Following interpolation, eye-tracking data was smoothed with a small low-pass Butterworth filter. Such post-processing allowed us to reach a $92.90\% \pm 5.45\%$ quota of usable data. BioPac physiological signals, collected at 2,000 Hz, were downsampled to match the frequency of vehicular and eye-tracking measurements (60 Hz), which is close to that of commercial fitness monitors, representing a more realistic approach for on-board deployment.

G. DATASET

The dataset includes 12 features for 162,800 samples, each one labeled as a distracted or not-distracted state (in a 50%-50% proportion, as per the data collection procedure).

TABLE 2. Data types.

Data type	Features
Vehicular driving data (D)	Speed X
	Speed Y
	Acceleration X
	Acceleration Y
	Steering angle
	Steering speed
	Steering torque
	Accelerator
	Head heading
	Head pitch
	Head roll
Eye-tracker data (E)	Eyelid Opening
	Left Eyelid Opening
	Right Eyelid Opening
	Robust Eyelid Opening
	Left Robust Eyelid Opening
	Right Robust Eyelid Opening
	Pupil Diameter
	Left Pupil Diameter
	Right Pupil Diameter
	Estimated Eye Position X
	Estimated Eye Position Y
	Estimated Eye Position Z
Physiological data (P)	Electrodermal activity (EDA)
	Heart rate (HR)

Featured data types are reported in Table 2, where X and Y are used for features representing spatial coordinates. As for eye-tracking data, eye coordinates positions X and Y were processed to create a new categorical feature that identifies the object on which the eye's gaze focused. The identified categories include left side, left mirror, road, dashboard, steering wheel, rear mirror, right mirror, right side, outside, and eye-sight not recognized. Categorical values were then processed with OneHotEncoding [66], so the single eye position identification feature was shaped into a sparse matrix consisting of 10 new features.

As a peculiar choice, we decided to take an end-to-end ML approach, considering as sample to be classified a whole multivariate timeseries in a given length time window (e.g., 1 second), thus using all the raw signals, as in state-of-the-art applications of deep learning models. Other state-of-the-art studies (e.g., Misra et al. [11]) use data points (sampled data plus a statistical synthesis of the values in the interval), instead.

For targeting a real-world vehicular application, we applied an overlapping moving windowing technique to the training and validation data, initially employing a stride of 10% of the total data points for the given window size. For simplicity, and given the very similar results obtained, testing windows were finally separated without any overlap, thereby generating multiple independent test samples.

As anticipated in the Related work section, we partitioned the dataset into three distinct subsets, selecting 70% of the users for training (30), 15% for validation (6), and 15% for testing (6). Other state-of-the-art studies (Gjoreski et al. [16],

Misra et al. [11]) have not adopted this solution. But, in view of a real-world system deployment, having the same users in the training, validation and testing sets may lead to an over-optimistic assessment, since it would assume that the DDD system would have been trained also on data from its actual users, before its deployment (or after-sale fine-tuning would be needed). Thus, we decided for a between-subject design, to ensure integrity and generalizability of the model across different subjects.

IV. MODEL TUNING

For our analysis, we optimized and tested various state-of-the-art ML models dedicated to timeseries classification to assess their performance in the task. This section briefly introduces them and describes their hyperparameter tuning.

A. 1D-CNN

1D Convolutional Neural Networks (1D-CNN) [67], [68] are well-suited for timeseries analysis as they are able to capture temporal features through local connectivity and shared weights, which makes them robust to shifts and suited to handle multivariate data. 1D convolution is performed by 2D kernels that aggregate values on the temporal and on the channel axes. For simplicity, Figure 4 shows an example with a 1D signal (i.e., single input channel).

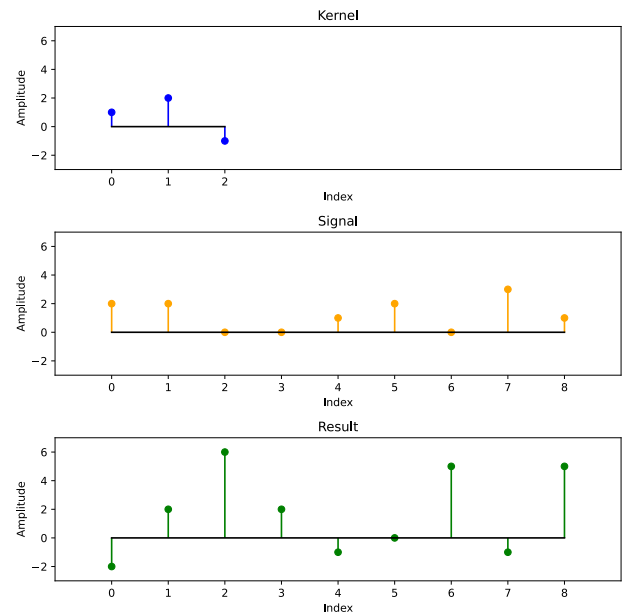


FIGURE 4. 1D convolution with a 1D input signal.

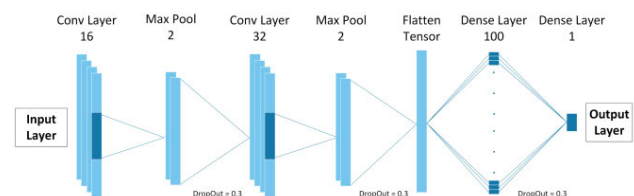


FIGURE 5. 1D-CNN model architecture.

To optimize the network, we performed a grid search exploring various kernel sizes (3, 5, 7), batch sizes (16, 32, 64, 128, 256), learning rates (10^{-4} , 10^{-3}), dropout rates (0.3, 0.5, 0.7), and number of kernels in the two convolutional layers (16-32, 24-48, 24-48-96, 64-128). The combination that resulted in the best accuracy was a kernel size of 3, batch size of 32, learning rate of 10^{-3} , dropout rate of 0.3, and 16-32 kernels (Figure 5). Our network uses the ReLU activation function, valid padding, and has a two-layer dense head, with the final sigmoid activation for extracting the distraction class probability.

B. MINIROCKET

Minimally random convolutional kernel transform (MiniRocket) [69] is a very training-efficient model exploiting the dilated convolution mechanism [70] to especially deal with long sequences. The MiniRocket transform generates features from timeseries data using convolutional kernels with predetermined weights, creating feature vectors through a proportion of positive values (PPV) pooling (Figure 6). Such features are then fed into a linear classifier, a Ridge classifier in our study (Figure 7). This approach minimizes hyperparameter tuning, which allows limiting the grid search to the number of kernels (we tested: 420, 504, 1008, 2688) and max number of dilations per kernel (5, 6, 12, 32). We obtained the best validation results for 420 kernels and 5 max dilations. MiniRocket involves just one convolutional layer (but with a large number of combinations of kernels and dilation, i.e., extracted feature vectors). However, we consider it a deep learning model, as it combines the convolutional layer with a shallow ML model such as the Ridge classifier.

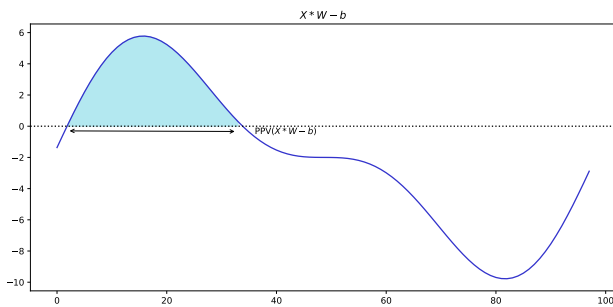


FIGURE 6. Example of PPV computation for a feature vector.

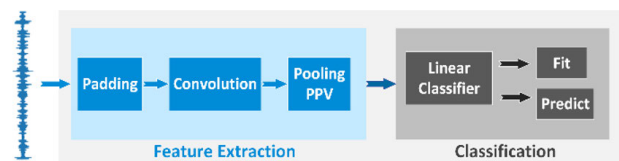


FIGURE 7. MiniRocket model architecture.

C. WAVENET

Dilated convolutions are largely exploited also by WaveNet [71], a deep model designed to generate audio waveforms, and now widely used in several applications thanks to its

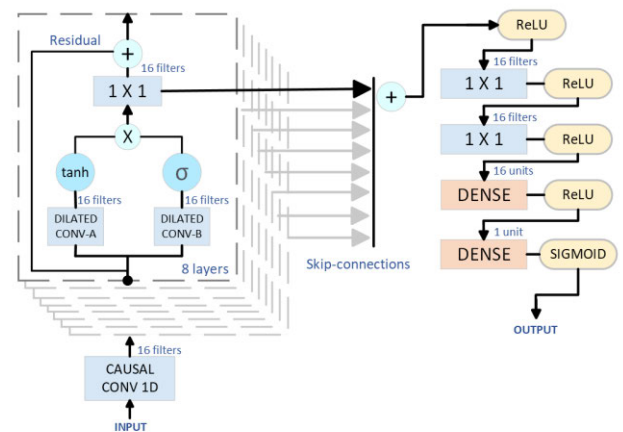


FIGURE 8. WaveNet model architecture.

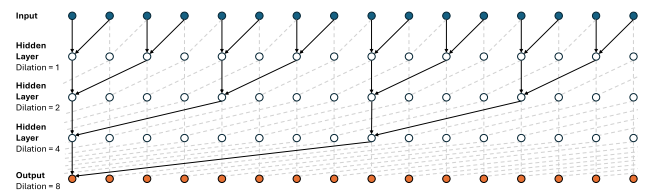


FIGURE 9. Stack of dilated convolutional layers.

ability to effectively capture long-range temporal dependencies, which allows it to efficiently process (long) timeseries. The architecture starts with a causal convolution and includes a stack of residual blocks with increasing dilation rates (Figure 8). The dilation mechanism allows to rapidly aggregate information from segments of a time series also with small, computationally efficient kernel sizes (size 2 in Figure 9).

Skip connections are output from each residual layer to create a vector, which is processed in the output stack. This stack includes cross-channel convolution and dense layers for the final class probability computation. Our grid search involved learning rates (10^{-4} , 10^{-3}), number of filters (both in blocks and head) (16, 32, 64), number of units in the head's dense layer (16, 32, 64), number of stacked layers (7, 8, 9), and batch size (32, 64, 128), finding the best values for learning rate 10^{-4} , filters 16, units 16, layers 8, and batch size 32 (Figure 8).

D. TRANSFORMERS

Transformers [72] have been used for timeseries classification, leveraging the well-known attention mechanism. We employed TensorFlow's implementation, where a transformer block with multiple attention heads enables the model to focus on different aspects of the data. Specifically, we utilized the MultiHeadAttention layer with 8 attention heads and embeddings' dimension of 128, followed by a feed-forward network (ffn) with 512 units. The model architecture consists of a decoder with 4 transformer blocks, each one containing multi-head attention (Figure 10), layer normalization, and a dropout mechanism for regularization. Positional encoding

was added to account for the sequence order in the time-series data. Finally, after global average pooling, a dense layer with sigmoid activation is used for binary classification (Figure 11). The parameters were computed with a grid search conducted on number of attention heads (4, 8, 16), embedding dimension size (64, 128, 256) and feedforward network size (128, 256, 512). The values for the best configuration are reported in Figure 10.

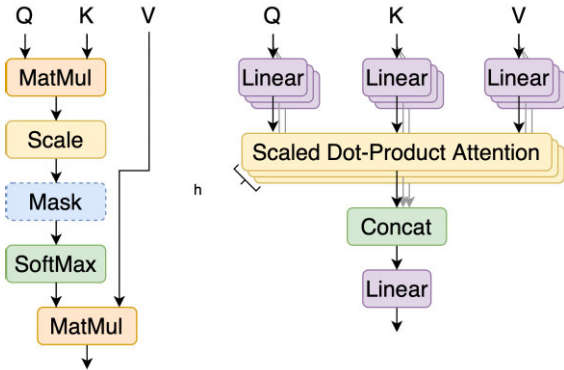


FIGURE 10. Scaled dot product attention (left) and multi-head attention (right).

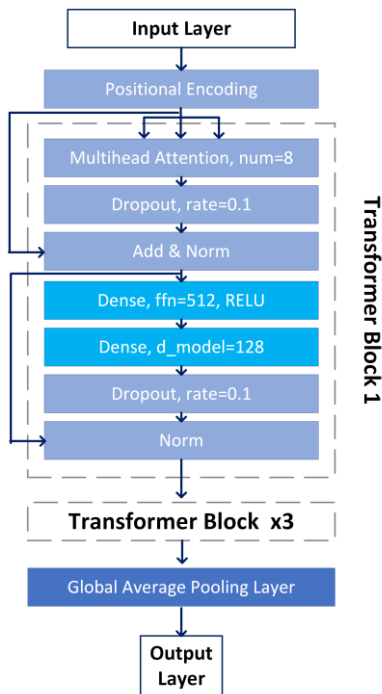


FIGURE 11. Transformer model architecture.

E. RESNET-18

Residual Networks (ResNets) [73] are popular in image recognition, thanks to their skip connections (Figure 12), which allow the building of deep hierarchical structures by mitigating the vanishing gradient problem. In adapting ResNet for timeseries data, we configured a 1D ResNet-18 version. Each residual block of the model consists of two

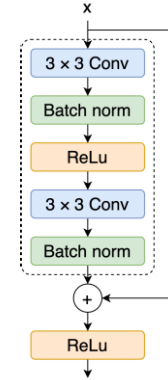


FIGURE 12. ResNet block.

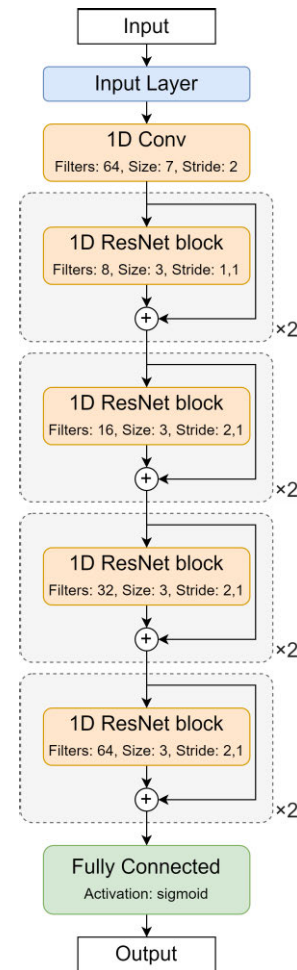


FIGURE 13. ResNet-18 model architecture.

1D convolutional layers followed by batch normalization and ReLU activation (Figure 12), with an optional convolutional shortcut if the input and output dimensions differ. The network in Figure 13 begins with an initial convolution, followed by four groups of residual blocks with increasing number of filters (8, 16, 32, and 64) and occasional downsampling through stride adjustments. After the final block, a global average pooling layer eliminates the temporal dimension, and a fully connected layer with a sigmoid activation outputs the

binary classification. The model is compiled using the Adam optimizer and binary cross-entropy loss. Grid search involves kernel size (3, 5, 7), finding the best option of 3.

V. EXPERIMENTAL RESULTS

This section presents and discusses our experimental results, which we compare with two key reference works in the area, namely Gjoreski et al. [16] and, particularly, Misra et al. [11]. We must preliminarily highlight that the three works (including ours) use three different proprietary datasets, which makes the comparison arbitrary. However, the goal of our work is to advance the state-of-the-art by answering the research questions proposed in the Introduction, trying to assess different experimental methods and ML models, also leveraging the most relevant knowledge and results published in the literature, such as the two above-mentioned papers.

A. 20-SECONDS WINDOW SIZE

As a first experiment, we used the same time-window size as Gjoreski et al. [16], i.e., 20 seconds. Table 3 reports a comparison of their experimental results with ours, first on the same set of signals (i.e., eye-tracker (E) + physiological (P)) then adding also driving (or vehicular) data (D), which is quite straightforward for a system targeted at on-board deployment. Considering shallow learning techniques, such as RF and eXtreme Gradient Boost (XGB), our RF performed worse than the same model in [16] (62% vs 67%), and XGB alike (69% vs 72%), which allows us to argue that our dataset is not intrinsically easier than that in [16]. On the other hand, our investigated deep learning models (i.e., 1D-CNN, MiniRocket, WaveNet, Transformer, and ResNet-18) perform substantially better than those in [16], reaching the highest F1 score with the 1D-CNN (82%).

TABLE 3. F1 score for 20 second time-windows (D = driving, E = eye-tracker, P = physiological data).

Dataset	Classifier	F1 score (%)
Gjoreski et al. [16] (E+P)	RF	67
	GB	73
	XGB	72
	eLSTM	67
	STRNet	67
Ours (E+P)	RF	62
	XGB	69
	1D-CNN	82
	MiniRocket	71
	WaveNet	73
	Transformer	81
	ResNet-18	81
Ours (D+E+P)	RF	74
	XGB	79
	1D-CNN	85
	MiniRocket	95
	WaveNet	85
	Transformer	84
	ResNet-18	91

These results align with established literature (e.g., [74]), which highlights that shallow models work well with smaller

datasets - particularly having a limited number of features, that are frequently selected through accurate manual feature engineering. DL models, on the other hand, can have a much larger number of learnable parameters, and are thus better suited to process high-complexity data, being able to perform and optimize feature extraction directly from raw signals, through end-to-end ML training. As we will see from the last group of results in Table 3, this also allows us to effectively process the vehicular signals, further boosting performance. Moreover, differently from the shallow learning models, all the proposed DL models are dedicated to timeseries (i.e., they intrinsically exploit the sequential nature of the input data) [75].

Considering our whole dataset (i.e., including vehicular driving signals beside eye-tracking and physiological data (D+E+P group in Table 3), MiniRocket outperforms all the other models, achieving an outstanding F1 score of 95%, followed by ResNet-18 (91%). This outcome confirms the known validity of the model (e.g., [76]) but is quite surprising given the fact that MiniRocket uses pre-determined convolutional weights, requiring very limited training time, which is essentially due to the linear classifier head. Accordingly, the MiniRocket's much lower performance on the reduced dataset (71% in E+P vs 95% in D+E+P) could be attributed to the deterministic, non-learnable nature of its convolutional kernels, which penalizes its performance in challenging datasets. This limit also raises concerns about the performance of this model in further and more various driving scenarios than those tested in the present experiment.

To a lesser extent than for MiniRocket, however performance of all our models has a significant surge in the complete dataset, which strongly hints at the importance of driving data for tackling the task. Particularly, we argue that the significantly lower F1 score in the (E+P) dataset than in the whole dataset could be due to the larger differences between subjects for physiological and eye-tracker signals rather than for vehicular signals. In fact, as we will discuss later, our results concern a test set with different users than the training set, while [16] does not specify such differentiation.

B. WINDOW SIZE REDUCTION

These results spurred us to investigate a reduction of the time window size, in view of an on-board deployable, real-time system, which needs much lower latencies. We thus consider samples of 20, 5, 2, 1, and 0.5 seconds, corresponding to 1,200, 300, 120, 60, and 30 data frames, respectively, within the 60 Hz dataset.

Figure 14 shows the performance of our ML algorithms with the different window sizes. Results indicate a decrease, which smooths down at around 1s. The chart also shows that ResNet-18 (and 1D-CNN alike, but with slightly worse results) has more consistent performance than the others across the different sizes, which may suggest its utilization also in a multi-resolution system. The performance loss is limited for the ResNet-18, which achieves a 78.04% F1

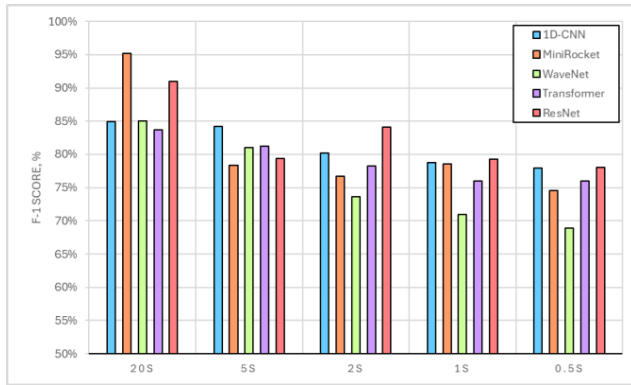


FIGURE 14. F1 scores for different window sizes.

score with a 0.5-time window (not much lower than 79.32% obtained with 1s) and also for the 1D-CNN (78.71% for 1s, vs 77.92% for 0.5s). This result is very significant, as it suggests the feasibility of a real-time system, processing a short time span at each shot.

C. CROSS-USER GENERALIZATION

A third experiment concerned the comparison with Misra et al. [11], who achieved a remarkable accuracy of almost 99% (on a D+E+P dataset). They use an RF model which samples the input signals every second and also incorporates four secondary features computed in the one-second window, namely: longitudinal velocity, standard deviation of the steering wheel angle, standard deviation of lateral position, and mean of the steering error (i.e., the deviation between the expected steering movement and the one performed by the driver). In contrast, our RF model achieves an accuracy of 78.15% and ResNet-18 of 83.00% (Table 4).

We argue that such a performance gap can be attributed to the different dataset split approaches employed. While Misra et al. [11] trained and tested their models on a mixed dataset (i.e., the training, validation, and testing subsets were split after random shuffling the samples from all the users), we opted for a stricter approach by splitting the dataset between subjects, reserving 30 users for training, 6 other users for validation and 6 others for testing. This is because we aim to evaluate the generalization capabilities of the models across new, unseen subjects [16], [77].

In order to give a novel, quantitative measure of the overfitting on the dataset subjects, Table 4 also report the results we obtained in an approach without generalization, (accuracy of 97.84% and 99.45%, for RF and ResNet-18, respectively),

that are comparable with those in Misra et al. [11]. This approach implies training a system with data also from its final users, which may lead to an over-optimistic performance assessment in view of a real-world deployment.

Extending the comparison to the different data types, Table 4 shows that the performance of our models always benefits from the presence of D and E data, which is not the case in [11], where the best result is obtained by processing physiological data only. The between and within-subject design comparison clearly shows that P signals alone are much more prone to inter-individual variability. RF and ResNet-18 achieve similar results on D and on E separately, both with and without generalization. Combining D and E generally increases the models' performance, while adding P to D or E causes a small drop in accuracy, which corresponds to the problematic collection of P data, as reported in subsection III-F (Signal post-processing).

The introduction of secondary features (as in [11]) leverages domain knowledge to enhance the input, while the deep-learning approach we follow synthesizes features by directly processing the raw input signals as part of the model training process. Full timeseries processing preserves the temporal dependencies and exploits the sequential nature of timeseries data, allowing models to learn complex patterns and interactions over time. This approach requires more sophisticated, deep learning models, such as 1D-CNN, ResNet-18 or MiniRocket, which are less likely to miss important temporal information. The ability to recognize complex patterns and still avoid overfitting appears from the superior performance of ResNet-18 with respect to RF in all the tasks with our dataset.

D. SIGNAL TYPES

A key factor for on-board system deployment concerns the type of signals to be processed. Particularly, vehicular data are easily accessible to original equipment manufacturers (OEMs), while collecting the others is more expensive and invasive for the driver. On the other hand, vehicular signals alone may not be sufficient to reach the necessary accuracy. We thus analyzed several combinations of data types (D, E and P), considering different types of deep learning models as well. As time-window size, we chose 0.5 s. This reduces performance compared to Table 4 (1 s), but can be considered a stress test, halving the system latency, which may make the difference, especially at high vehicle's speed.

TABLE 4. Influence of generalization, signal type and timeseries-oriented neural models on accuracy for 1s windows.

	D+P+E	D+E	D+P	E+P	D	E	P
RF in Misra et al. [11] (without generalization)	98.75%	95.08%	98.13%	99.37%	71.57%	96.32%	99.58%
RF with generalization	78.15%	76.16%	69.92%	69.40%	71.42%	71.05%	66.37%
ResNet-18 with generalization	83.00%	80.68%	77.90%	76.05%	78.60%	78.19%	69.01%
RF without generalization	97.84%	94.70%	93.65%	93.47%	94.98%	94.22%	86.46%
ResNet-18 without generalization	99.45%	99.06%	97.85%	97.09%	97.67%	97.28%	92.09%

TABLE 5. Results for Different types of input data for 0.5 s window.

Dataset	Classifier	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Training time (s)	Model size (KB)
D	1D-CNN	72.03	74.44	67.12	70.60	201	329
	MiniRocket	67.51	68.00	66.11	67.04	18	86
	WaveNet	69.77	70.23	69.77	69.99	85	793
	Transformer	70.69	74.50	65.72	69.84	119	44
	ResNet-18	72.35	67.79	75.62	71.49	219	1,100
D+E	1D-CNN	75.12	75.36	71.57	73.41	155	339
	MiniRocket	72.36	70.99	74.54	72.72	13	92
	WaveNet	71.95	72.01	71.89	71.95	89	833
	Transformer	74.68	72.40	71.30	71.84	120	60
	ResNet-18	75.40	75.23	75.05	75.14	218	1,159
D+P	1D-CNN	71.93	74.30	63.98	68.76	98	331
	MiniRocket	69.04	65.40	72.94	68.96	12	84
	WaveNet	69.03	69.03	70.75	69.88	90	997
	Transformer	71.20	72.32	69.34	70.79	94	49
	ResNet-18	70.75	62.28	76.27	68.57	240	1,130
E	1D-CNN	69.31	71.98	67.71	69.78	73	310
	MiniRocket	67.68	66.40	68.33	67.35	19	101
	WaveNet	69.53	70.13	69.84	69.99	87	803
	Transformer	71.12	73.28	69.23	71.20	129	47
	ResNet-18	71.99	63.00	78.77	70.01	239	950
E+P	1D-CNN	68.17	70.48	63.31	66.70	95	311
	MiniRocket	65.42	66.58	64.52	65.53	12	43
	WaveNet	66.93	67.06	66.93	67.00	91	829
	Transformer	69.25	68.94	68.82	68.88	128	56
	ResNet-18	68.36	70.07	67.05	68.53	187	1,061
D+E+P	1D-CNN	77.13	76.58	78.55	77.55	188	340
	MiniRocket	75.10	75.88	74.01	74.93	20	188
	WaveNet	69.94	71.22	69.90	70.55	97	835
	Transformer	75.34	74.55	75.43	74.99	120	69
	ResNet-18	78.08	74.38	81.61	77.83	236	1,233

A comprehensive analysis, reported in Table 5, confirms the fundamental importance of vehicular signals. With our experimental data, the added value of P signals looks limited, as we have commented in the previous sub-section for a slightly larger time-window. We verified that the P signal performance is not penalized by our 60 Hz sub-sampling, by assessing performance also at 120, 180, and 240 Hz, according to the Biopac guide [78] (page 8), obtaining similar results. However, notably, the best performance is obtained by combining all the signal types together (D+E+P), which indicates that the investigated models are able to identify additional general patterns while learning to be robust also to possible sensor failure, as it happened for P. The ResNet-18 model trained on the full dataset achieves the highest accuracy of 78.08%, with the 1D-CNN reaching a close value of 77.13%.

The (E+P) dataset is particularly important as in the automated driving condition where vehicular signals do not depend on the driver activity. At high levels of driving automation, distraction detection is very useful to achieve an effective take-over request (TOR) [79], which is issued for instance in case of an ADF system failure or exit from its

operational design domain (ODD). Results from the (E + P) dataset, where Transformer achieves a 69.25% accuracy, indicate a lot of room for research in this direction, particularly to mitigate the inter-individual variability issue.

Considering a more comprehensive metric, such as the area under curve (AUC) for the precision-recall curve, we obtain results substantially consistent with the F1-score (e.g., 0.73 and 0.74 in D for 1D-CNN and ResNet, respectively; and 0.85 and 0.86 in D+E+P for 1D-CNN and ResNet, respectively). Figure 15 presents the precision-recall curves for ResNet-18 in the different data type combinations (always with a 0.5s window). The chart confirms the overall superiority of the complete dataset, followed by the D+E combination, while addition of physiological data looks useful at low recall. Significantly, the complete system seems to be able to keep relatively high precision at very high recall levels.

As predictable, given its well known training efficiency [70], MiniRocket has the lowest training times, between 12 and 20 seconds. But the training time is limited for all the other models as well (up to less than 4 minutes). Inference times depend on the target platform, and we will

discuss them later in the sub-section dedicated to embedded system deployability.

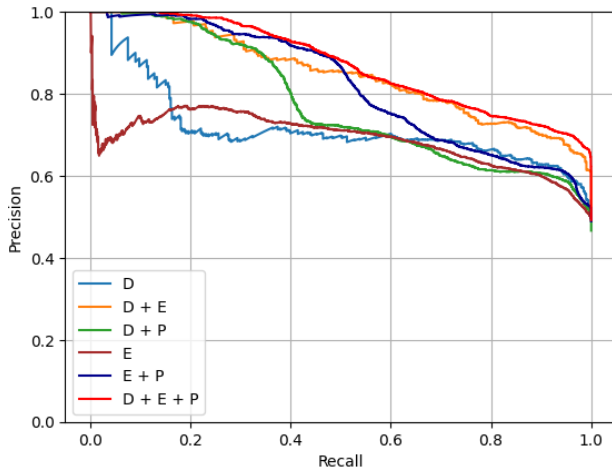


FIGURE 15. ResNet-18 precision-recall curves for different types of input data for the 0.5 s. window.

The size of the MiniRocket and Transformer models is very small (in the order of tens of KB), while 1D-CNN, ResNet-18, and, overall, WaveNet are one order of magnitude larger.

E. ERROR ANALYSIS

Table 6 reports the confusion matrices of the 0.5s ResNet-18 for two sample users for whom the system performs quite less than the average (accuracy of 71.67% and 76.95%, respectively). Figure 16 shows the evolution of the predictions during their 180 s test. The first sample clearly indicates the difficulty of our system in the transition phase (i.e., at the beginning of the TQT, which happens in the middle of the timeline). This is a clear challenge, essentially because actual distraction may not start immediately at the start of the task. On the other hand, the second user shows distracted behavior, at least according to our model, also without the cognitive task. For this user the system is very sensitive, but also very imprecise. These examples highlight the significant difference in subjects' responses and attitudes, and hints at future research work on better feature extraction, user segmentation, and custom fine-tuning of ML models. In this sense, it is useful to add that we tried to cluster the test users according to their metadata (e.g., total real-world driven km, age, sex), but could not achieve better results. This is worth more investigation in future work, since these factors may affect acceptance and effectiveness of ADAS (e.g., Son et al. [80]).

F. EXPLAINABILITY

We then employed Shapley Additive exPlanations (SHAP) analysis to interpret the predictions of our best-performing model, namely the ResNet-18 trained on the full dataset, applied to the most critical, 0.5-s time window size. SHAP values provide a unified measure to understand the contribution of each feature to the model's output, offering both local and global interpretability (i.e., at sample and whole

TABLE 6. Prediction confusion matrix for two sample users.

Prediction \ Label	User 1		User 2	
	Positive	Negative	Positive	Negative
Positive	33.30%	16.70%	49.94%	0.06%
Negative	11.23%	38.37%	22.99%	27.01%

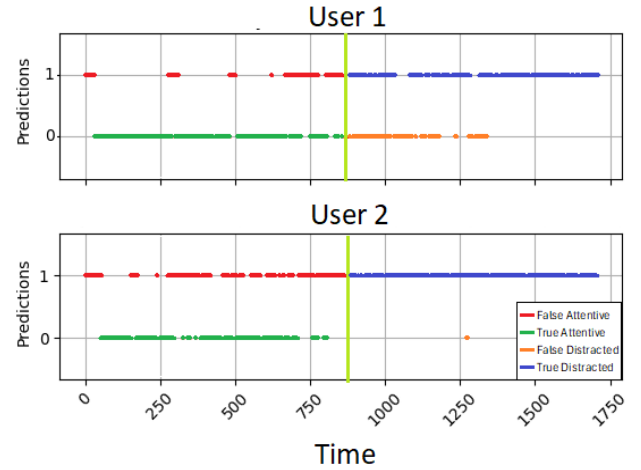


FIGURE 16. ResNet-18 binary predictions over time for two sample users.

dataset level, respectively) [81]. SHAP is a well-established, game theoretic approach to explain the output of any machine learning model, without knowing its internals (black-box explainability) [82].

Figure 17 shows the ResNet-18's top 10 features in terms of mean absolute SHAP values. Such values provide a global measure of feature importance and are thus useful to understand what features are more relevant, on average, for the decision making process of the analyzed model. It can be seen that six eye-tracker and four vehicular data have the largest values. Particularly, the strongest contribution to the model's predictive power comes from the accelerator signal, the estimated eye position, the head pitch and roll, and the lateral velocity. These findings align with and deepen the results presented in Table 5, where vehicular and eye-tracker data allowed reaching an accuracy of more than 75%. The SHAP top 10 feature ranking (Figure 17) does not include the HR and EDA physiological signals, consistently with the fact that P signals seem to contribute to overall performance less than the other two types of signals (Table 5). A specific result of our experiment analysis is given by the importance of the accelerator signal as a strong indicator of a driver's cognitive "presence". Comparing the distribution of the normalized signal values under the two conditions confirms that distraction leads to a much more uniform driver behavior in terms of pedal activity (Figure 18). The average and standard deviation values are 0.42 (0.19) vs. 0.40 (0.13), in the non-distraction and distraction cases, respectively.

Going to a local analysis, Figures 19 and 20 show, as an example, the SHAP values for a correct "attention" and "distraction" sample classification, respectively. The charts confirm that signals such as Accelerator, estimated eye

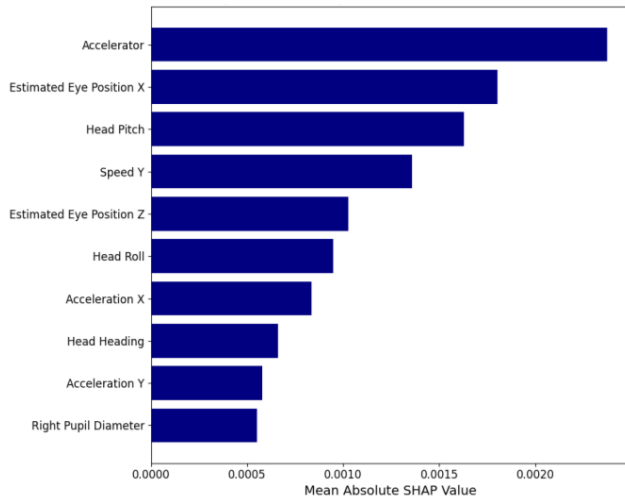


FIGURE 17. Top 10 features by highest absolute SHAP value for the ResNet-18 model on the full dataset.

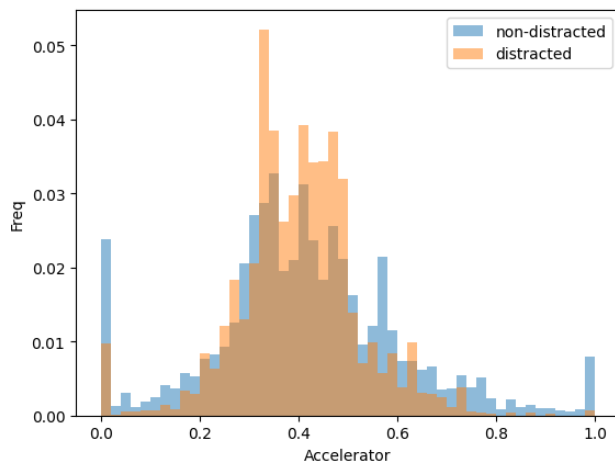


FIGURE 18. Accelerator signal profile in the two conditions.

position, speed and accelerations (y and x) are main motivators for the decision. Looking at their specific (normalized) values, it is interesting to note, as above, that attention seems related to control (higher values for lateral speed, accelerator and steering torque), opposed to distraction (longitudinal acceleration, steering angle different from 0, but torque almost null). Eye position and head pitch seem to be more relevant to classify distraction than attention. Physiological signals (HR and EDA) provide a small, decision-reinforcing contribution.

Misra et al. [11] made a feature importance analysis on their ExtraTree classifier, and highlighted the relevance of eye-tracking data. In our case, the SHAP analysis reveals that the largest contribution comes not only from eye-tracker signals, but also from the vehicular ones, which is a novelty in literature and an important outcome of our experiment, since such data is easily available in vehicles. We argue that this advancement is due to our proposal of processing vehicular signals with deep learning models specifically dedicated to time series processing. From an industrial perspective, this

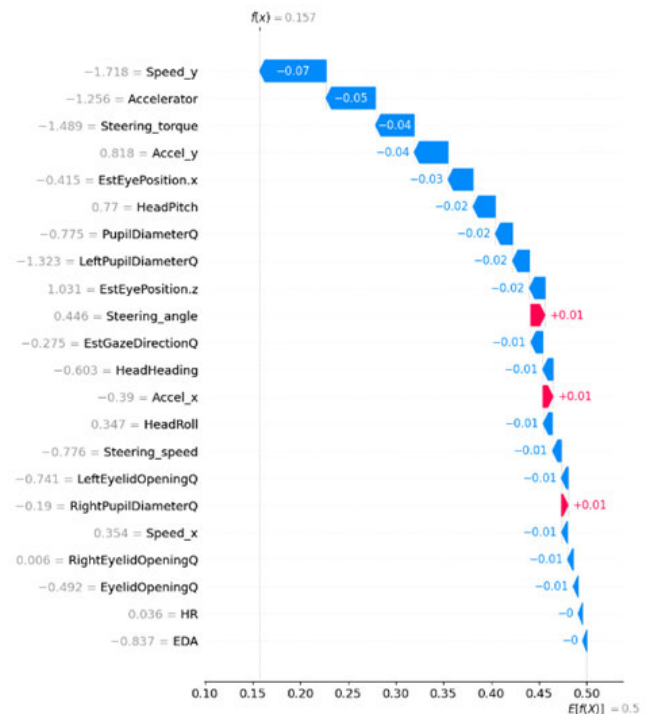


FIGURE 19. SHAP waterfall diagram of a correct "attention" classification.

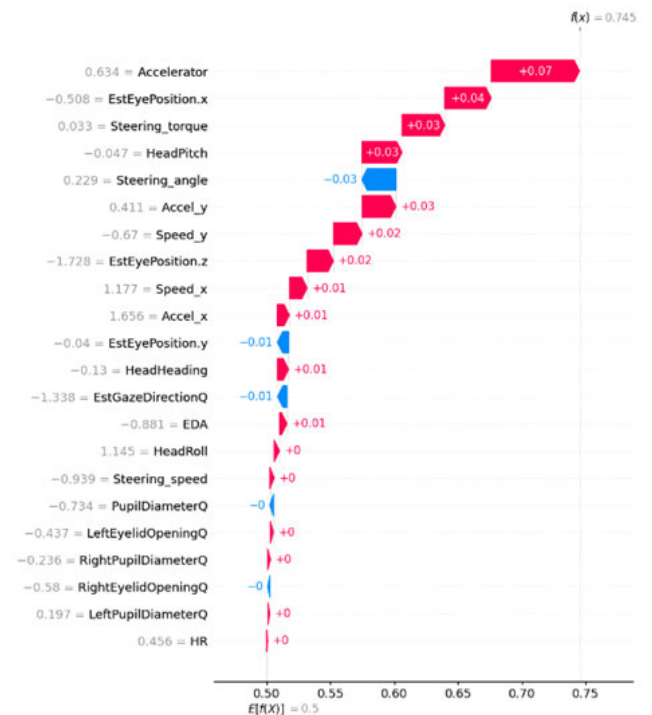


FIGURE 20. SHAP waterfall diagram of a correct "distraction" classification.

can contribute to reduce of dependency on external (i.e., non-vehicular) systems, that are costly and invasive. Beyond the technical detail of the SHAP analysis, knowing that vehicular signals are an important factor for the decision making process by DL models in the driver cognitive distraction

detection task is an arguably relevant novelty for non-expert stakeholders as well.

TABLE 7. Single user fine-tuning and user-extended training methods.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Original ResNet-18	78.08	74.38	82.08	77.97
Single user fine-tuning	98.52 (+20.44)	99.19 (+24.81)	97.78 (+16.17)	98.43 (+20.60)
User-extended training	96.04 (+17.96)	94.78 (+20.40)	95.86 (+14.25)	95.32 (+17.49)

G. FINE-TUNING

In our experiment, the DDD has a user-generalizable performance of less than 80%. However, better results could be achieved with subsequent user-specific adaptation. To assess adaptation to an unseen user, we explored two methods: a single-user fine-tuning and a user-extended fine-tuning. In both cases, we adapt the fully trained ResNet-18 model by using 50% of the target user samples for the fine-tuning, while leaving the other 50% for the testing.

In the single-user case, we fine-tuned the model exclusively on a single new user (taken from our test set). Such fine-tuning was performed for 30 epochs with a reduced learning rate (10^{-4} , one decade less than the original), resulting in significantly improved metrics (Table 7, presenting the average over all the test users). However, this approach causes the model to lose its generalization capabilities, as it appears from a $\sim 15\%$ loss in all the metrics, averaging over all the other test users.

In the user-extended fine-tuning experiment, instead, we incorporated the new user into the training set, fine-tuning the model using a 10^{-4} learning rate, but for only 10 additional epochs. We iterated the approach for each user in the test set, achieving a 17% average improvement in F1 score (Table 7). While this result is lower than that of the single-user case, this second method maintains the model's ability to generalize, showing for the other users in the test set a $\pm 2\%$ performance variation compared to the original model.

H. DEPLOYABILITY

In view of on-board deployment, Tables 8 and 9 report specific metrics (namely: model size, inference time, power consumption, and energy consumption per sample) measured on a system prototype deployed on a Raspberry Pi 5 microcontroller and a Jetson Nano microcomputer. The Raspberry Pi 5 has compact dimensions and an affordable price, and can execute deep learning tasks efficiently, through a quad-core ARM Cortex A76 processor. The processor runs at a speed of 2.4 GHz, and the device includes 8 GB of RAM and a 32 GB SD card for storage and booting. The Jetson Nano, on the

other hand, includes a quad-core ARM Cortex A57 processor that runs at a speed of 1.43 GHz, a NVIDIA Maxwell GPU, 4GB of RAM and a 32 GB SD card. These ranges of values are significant for automotive deployment, while other optimizations would be needed for boards having lower memory and computational capacity (e.g. STM 32), that we leave as future work.

To measure power metrics on the Raspberry Pi 5, we employed a JT-UM120 USB multimeter connected between the power supply and the device to track power variations. For the Jetson Nano, instead, we used the “jtop” integrated NVIDIA software. Energy consumption is computed on a single 0.5s time window. We have limited our analysis to the ResNet-18 and 1D-CNN models, because their ML performance and memory footprint make them particularly promising for embedded deployment. Results in Tables 8 and 9 show that the memory footprint of the models is under the 0.4 MB value, which is a typical limit for microcontrollers' ROM. Due to their higher complexity, the ResNet-18 models are 3.3x bigger than the 1D-CNN models. In all cases, the inference time is extremely short, making it compatible with real-time execution. We highlight that we reduced the latency by two orders of magnitude by converting the model from the .h5 to the Open Neural Network Exchange (ONNX) format, which has a significant impact also in terms of energy consumption. The difference in execution time between the Raspberry Pi 5 and the Jetson Nano can be attributed to the fact that the Raspberry Pi 5's CPU is a newer-generation quad-core ARM Cortex A76 processor, offering higher clock speeds (2.4 GHz) and improved architectural efficiency compared to the older quad-core ARM Cortex A57 processor in the Jetson Nano (1.43 GHz). Cross-referencing data from Table 5 and Tables 8 and 9, we note that while ResNet 18 has a slightly better classification performance, its energy consumption is significantly higher than 1D-CNN, which highlights an important trade-off to be considered by designers.

TABLE 8. Deployability metrics on a raspberry Pi 5.

Dataset	Classifier	Model size onnx (KB)	Execution Time (μ s)	Power Cons. (mW)	Energy Cons. (μ J)
D	1D-CNN	92.3	45	850	38
	ResNet-18	306.9	250	1,100	277
D+E+P	1D-CNN	97.1	50	950	48
	ResNet-18	350.4	270	1,200	314

TABLE 9. Deployability metrics on a jetson nano (CPU).

Dataset	Classifier	Model size onnx (KB)	Execution Time (μ s)	Power Cons. (mW)	Energy Cons. (μ J)
D	1D-CNN	92.3	175	1,550	271
	ResNet-18	306.9	860	1,600	1,376
D+E+P	1D-CNN	97.1	195	1,650	322
	ResNet-18	350.4	890	1,650	1,469

Notably, for the Jetson Nano, Table 9 reports results for the CPU, because the GPU had longer inference times and higher energy consumption. We attribute this to the overhead of utilizing the GPU and the inefficiency of the parallelism for small batches and lightweight tasks. Particularly, compared with the same inference task on CPU, the inference time and energy consumption on the GPU are $\sim 37\times$ and $\sim 9\times$ higher for 1D-CNN and ResNet-18, respectively.

VI. CONCLUSION AND FUTURE WORK

We have investigated the feasibility of a model of driver's cognitive distraction with performance suitable for on-board deployment. After a data collection campaign with 42 subjects undertaking the TQT in a driving simulator, we show that state of the art DL models dedicated to time series processing outperform shallow ML models, arguably because of their ability to optimize feature extraction directly from raw signals through end-to-end ML training. Results also show that using vehicular, eye-tracking, and physiological signals (sampled at a frequency similar to that of easily deployable commercial fitness monitors) allows achieving an accuracy of 78.08%, with a time window of just 0.5 s, using a ResNet-18 classifier. To the best of our knowledge, these are the first results reported in literature on cognitive distraction including vehicular signals and separating test users from training users (between-subject design). If test users are the same as train users, accuracy increases to 99.45%. We argue that the large gap between these two results is a clear indicator of the significance of the inter-individual variability issue. Through a SHAP AI explainability analysis, we showed that both vehicular signals and eye-tracker data are fundamental to ensure performance, while physiological signals are less important, even if still useful, in our dataset. The between and within-subject test design comparison showed that the physiological signals are much more prone to inter-individual variability, and they are also susceptible to data loss or signal degradation, which are fundamental factors to consider for commercial deployment, as they may require fine-tuning the system with data from the actual end-user. Our analysis has quantitatively measured the effect of inter-individual variability for all types of signals, demonstrating its relevance, and shown that DL models dedicated to timeseries processing are better able to generalize across users than shallow ML models. DL models demonstrate also robustness to sensor failures. With a focus on in-vehicle deployability, which is of key industrial interest, we measured also such metrics as model size, inference time, and energy consumption, demonstrating feasibility on a prototype embedded platform.

We believe that the original findings of this research provide significant advancement towards on-board deployment of robust cognitive distraction detection systems, able to work in real-time. They also indicate some key directions for future research. Overall, inter-individual variability must be carefully addressed. Unsupervised ML techniques may be explored for this, trying to automatically find patterns in larger quantities of unlabeled data, also considering the

objective difficulty in defining the ground truth for cognitive distraction (i.e., the driver is labeled as distracted or not depending on whether he/she is performing a distracting task, but this approach is somehow arbitrary). User and context adaptivity should be considered as well [83].

We have investigated the performance of an end-to-end ML approach. Results indicate that there is room for improvement, possibly by developing better models, especially for short time-windows. On the other hand, the NN could be inserted in a comprehensive system architecture, implementing expert knowledge, particularly to address criticalities, for instance the transition phase at the beginning of the distracting task (Figure 16). Real-time adaptivity could be considered as well.

Our study relies on a static simulation environment, that we set up to minimize the experimental factors and get timely response to our three fundamental research questions. In light of the obtained results, it is now important to apply the proposed research methodology, namely synthesizing: (i) time series-dedicated DL models, (ii) train and test user split, (iii) embedded deployability metrics, to more complex and dynamic driving simulation settings. Similarly, more experiments should be conducted under different conditions, including distraction tasks, driving scenarios, higher levels of automation, sensors (particularly for physiological measurements) and environmental noise, while also assessing user acceptance [84]. Finally, once effectiveness has been proven in the laboratory, extensive user tests should be performed on a fully equipped vehicle prototype, despite the higher costs and overall risks associated with real-world settings. For this, deployment on lower-resource devices should be investigated as well.

Our research has focused on cognitive distraction. Detecting cognitive distraction is quite different from visual and manual distraction, as discussed in the Introduction. So, while some of this paper's findings may be generalizable, specific research work should be done to better investigate this aspect.

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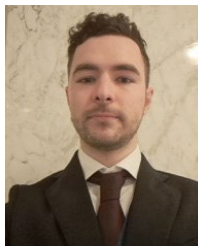
REFERENCES

- [1] A. Kashevnik, R. Shchedrin, C. Kaiser, and A. Stocker, "Driver distraction detection methods: A literature review and framework," *IEEE Access*, vol. 9, pp. 60063–60076, 2021, doi: [10.1109/ACCESS.2021.3073599](https://doi.org/10.1109/ACCESS.2021.3073599).
- [2] M. Doudou, A. Bouabdallah, and V. Berge-Cherfaoui, "Driver drowsiness measurement technologies: Current research, market solutions, and challenges," *Int. J. Intell. Transp. Syst. Res.*, vol. 18, no. 2, pp. 297–319, May 2020, doi: [10.1007/s13177-019-00199-w](https://doi.org/10.1007/s13177-019-00199-w).
- [3] T. Stewart, "Overview of motor vehicle crashes in 2020," U.S. Dept. Transp., Nat. Highway Traffic Safety Admin., Washington, DC, USA, Tech. Rep. DOT HS 813 266, Mar. 2022. Accessed: Aug. 5, 2024. [Online]. Available: <https://trid.trb.org/View/1922738Tech>

- [4] R. Fredriksson, M. G. Lenné, S. van Montfort, and C. Grover, "European NCAP program developments to address driver distraction, drowsiness and sudden sickness," *Frontiers Neuroergonomics*, vol. 2, Nov. 2021, Art. no. 786674, doi: [10.3389/frngo.2021.786674](https://doi.org/10.3389/frngo.2021.786674).
- [5] T. Ersal, H. J. A. Fuller, O. Tsimhoni, J. L. Stein, and H. K. Fathy, "Model-based analysis and classification of driver distraction under secondary tasks," *IEEE Trans. Intell. Transp. Syst.*, vol. 11, no. 3, pp. 692–701, Sep. 2010, doi: [10.1109/TITS.2010.2049741](https://doi.org/10.1109/TITS.2010.2049741).
- [6] M. A. Regan, C. Hallett, and C. P. Gordon, "Driver distraction and driver inattention: Definition, relationship and taxonomy," *Accident Anal. Prevention*, vol. 43, no. 5, pp. 1771–1781, Sep. 2011, doi: [10.1016/j.aap.2011.04.008](https://doi.org/10.1016/j.aap.2011.04.008).
- [7] M. A. Regan, J. D. Lee, and K. Young, *Driver Distraction: Theory, Effects, and Mitigation*. Boca Raton, FL, USA: CRC Press, 2008, doi: [10.1201/9781420007497](https://doi.org/10.1201/9781420007497).
- [8] R. Dewar and P. Olson, *Human Factors in Traffic Safety*, 2nd ed. Tucson, AZ, USA: Lawyers & Judges Pub. Co., 2007. Accessed: Aug. 5, 2024. [Online]. Available: <https://trid.trb.org/View/815003>
- [9] G. Li, G. Wang, Z. Guo, Q. Liu, X. Luo, B. Yuan, M. Li, and L. Yang, "Domain adaptive driver distraction detection based on partial feature alignment and confusion-minimized classification," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 9, pp. 11227–11240, Sep. 2024, doi: [10.1109/TITS.2024.3367665](https://doi.org/10.1109/TITS.2024.3367665).
- [10] V. Nagy, P. Földesi, and G. Istenes, "Area of interest tracking techniques for driving scenarios focusing on visual distraction detection," *Appl. Sci.*, vol. 14, no. 9, p. 3838, Apr. 2024, doi: [10.3390/app14093838](https://doi.org/10.3390/app14093838).
- [11] A. Misra, S. Samuel, S. Cao, and K. Shariatmadari, "Detection of driver cognitive distraction using machine learning methods," *IEEE Access*, vol. 11, pp. 18000–18012, 2023, doi: [10.1109/ACCESS.2023.3245122](https://doi.org/10.1109/ACCESS.2023.3245122).
- [12] P. D'Addario and B. Donmez, "The effect of cognitive distraction on perception-response time to unexpected abrupt and gradually onset roadway hazards," *Accident Anal. Prevention*, vol. 127, pp. 177–185, Jun. 2019, doi: [10.1016/j.aap.2019.03.003](https://doi.org/10.1016/j.aap.2019.03.003).
- [13] Y. Qiao, X. Yang, J. Wang, T. Si, and Q. Guo, "Driver cognitive distraction detection based on eye movement behavior and integration of multi-view space-channel feature," *Expert Syst. Appl.*, vol. 266, Mar. 2025, Art. no. 125975, doi: [10.1016/j.eswa.2024.125975](https://doi.org/10.1016/j.eswa.2024.125975).
- [14] B. A. Anderson, "Time to stop calling it attentional 'captur' and embrace a mechanistic understanding of attentional priority," *Vis. Cognition*, vol. 29, no. 9, pp. 537–540, Oct. 2021, doi: [10.1080/13506285.2021.1892894](https://doi.org/10.1080/13506285.2021.1892894).
- [15] M. A. Recarte and L. M. Nunes, "Mental workload while driving: Effects on visual search, discrimination, and decision making," *J. Experim. Psychol., Appl.*, vol. 9, no. 2, pp. 119–137, 2003, doi: [10.1037/1076-898x.9.2.119](https://doi.org/10.1037/1076-898x.9.2.119).
- [16] M. Gjoreski, M. Ž. Gams, M. Luštrek, P. Genc, J.-U. Garbas, and T. Hassan, "Machine learning and end-to-end deep learning for monitoring driver distractions from physiological and visual signals," *IEEE Access*, vol. 8, pp. 70590–70603, 2020, doi: [10.1109/ACCESS.2020.2986810](https://doi.org/10.1109/ACCESS.2020.2986810).
- [17] F. Tango and M. Botta, "Real-time detection system of driver distraction using machine learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 14, no. 2, pp. 894–905, Jun. 2013, doi: [10.1109/TITS.2013.2247760](https://doi.org/10.1109/TITS.2013.2247760).
- [18] A. Capello, M. Fresta, F. Bellotti, H. Haghighi, J. Hiller, S. Mozaffari, and R. Berta, "Exploiting big data for experiment reporting: The hi-drive collaborative research project case," *Sensors*, vol. 23, no. 18, p. 7866, Sep. 2023, doi: [10.3390/s23187866](https://doi.org/10.3390/s23187866).
- [19] A. Barisic, P. Sigris, S. Oliver, A. Sciarra, and M. Winckler, "Driver model for take-over-request in autonomous vehicles," in *Proc. Adjunct 31st ACM Conf. User Model., Adaptation Personalization*. New York, NY, USA: ACM, Jun. 2023, pp. 317–324, doi: [10.1145/3563359.3596994](https://doi.org/10.1145/3563359.3596994).
- [20] J. B. Manchon, M. Bueno, and J. Navarro, "Calibration of trust in automated driving: A matter of initial level of trust and automated driving style?" *Human Factors: J. Human Factors Ergonom. Soc.*, vol. 65, no. 8, pp. 1613–1629, Dec. 2023, doi: [10.1177/00187208211052804](https://doi.org/10.1177/00187208211052804).
- [21] R. Roscher, B. Bohn, M. F. Duarte, and J. Garcke, "Explainable machine learning for scientific insights and discoveries," *IEEE Access*, vol. 8, pp. 42200–42216, 2020, doi: [10.1109/ACCESS.2020.2976199](https://doi.org/10.1109/ACCESS.2020.2976199).
- [22] K. L. S. Kumar and M. K. J. Kannan, "A survey on driver monitoring system using computer vision techniques," in *Innovative Computing and Communications*, A. E. Hassanien, S. Anand, A. Jaiswal, and P. Kumar, Eds., Singapore: Springer, 2024, pp. 253–262, doi: [10.1007/978-981-97-3591-4_21](https://doi.org/10.1007/978-981-97-3591-4_21).
- [23] D. Crundall, M. Bains, P. Chapman, and G. Underwood, "Regulating conversation during driving: A problem for mobile telephones?" *Transp. Res. F, Traffic Psychol. Behav.*, vol. 8, no. 3, pp. 197–211, May 2005, doi: [10.1016/j.trf.2005.01.003](https://doi.org/10.1016/j.trf.2005.01.003).
- [24] T. T. Kubose, K. Bock, G. S. Dell, S. M. Garnsey, A. F. Kramer, and J. Mayhugh, "The effects of speech production and speech comprehension on simulated driving performance," *Appl. Cognit. Psychol.*, vol. 20, no. 1, pp. 43–63, 2006, doi: [10.1002/acp.1164](https://doi.org/10.1002/acp.1164).
- [25] A. Chaparro, J. M. Wood, and T. Carberry, "Effects of age and auditory and visual dual tasks on closed-road driving performance," *Optometry Vis. Sci.*, vol. 82, no. 8, pp. 747–754, Aug. 2005, doi: [10.1097/01.opx.0000174724.74957.45](https://doi.org/10.1097/01.opx.0000174724.74957.45).
- [26] W. J. Horrey, M. F. Lesch, and A. Garabet, "Dissociation between driving performance and drivers' subjective estimates of performance and workload in dual-task conditions," *J. Saf. Res.*, vol. 40, no. 1, pp. 7–12, Jan. 2009, doi: [10.1016/j.jsr.2008.10.011](https://doi.org/10.1016/j.jsr.2008.10.011).
- [27] N. Merat, A. H. Jamson, F. C. H. Lai, and O. Carsten, "Highly automated driving, secondary task performance, and driver state," *Hum. Factors, J. Hum. Factors Ergonom. Soc.*, vol. 54, no. 5, pp. 762–771, Oct. 2012, doi: [10.1177/0018720812442087](https://doi.org/10.1177/0018720812442087).
- [28] G. Castignani, T. Dermann, R. Frank, and T. Engel, "Driver behavior profiling using smartphones: A low-cost platform for driver monitoring," *IEEE Intell. Transp. Syst. Mag.*, vol. 7, no. 1, pp. 91–102, Spring. 2015, doi: [10.1109/TITS.2014.2328673](https://doi.org/10.1109/TITS.2014.2328673).
- [29] A. Aksjonov, P. Nedoma, V. Vodovozov, E. Petlenkov, and M. Herrmann, "Detection and evaluation of driver distraction using machine learning and fuzzy logic," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 6, pp. 2048–2059, Jun. 2019, doi: [10.1109/TITS.2018.2857222](https://doi.org/10.1109/TITS.2018.2857222).
- [30] D. L. Strayer, J. Turrill, J. M. Cooper, J. R. Coleman, N. Medeiros-Ward, and F. Biondi, "Assessing cognitive distraction in the automobile," *Hum. Factors, J. Hum. Factors Ergonom. Soc.*, vol. 57, no. 8, pp. 1300–1324, Dec. 2015, doi: [10.1177/0018720815575149](https://doi.org/10.1177/0018720815575149).
- [31] G. Li, W. Yan, S. Li, X. Qu, W. Chu, and D. Cao, "A temporal-spatial deep learning approach for driver distraction detection based on EEG signals," *IEEE Trans. Autom. Sci. Eng.*, vol. 19, no. 4, pp. 2665–2677, Oct. 2022, doi: [10.1109/TASE.2021.3088897](https://doi.org/10.1109/TASE.2021.3088897).
- [32] S. V. Deshmukh and O. Dehzangi, "ECG-based driver distraction identification using wavelet packet transform and discriminative kernel-based features," in *Proc. IEEE Int. Conf. Smart Comput. (SMARTCOMP)*, May 2017, pp. 1–7, doi: [10.1109/SMARTCOMP.2017.7947003](https://doi.org/10.1109/SMARTCOMP.2017.7947003).
- [33] X. Chen, J. Cheng, R. Song, Y. Liu, R. Ward, and Z. J. Wang, "Video-based heart rate measurement: Recent advances and future prospects," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 10, pp. 3600–3615, Oct. 2019, doi: [10.1109/TIM.2018.2879706](https://doi.org/10.1109/TIM.2018.2879706).
- [34] H. Xiao, T. Liu, Y. Sun, Y. Li, S. Zhao, and A. Avolio, "Remote photoplethysmography for heart rate measurement: A review," *Biomed. Signal Process. Control*, vol. 88, Feb. 2024, Art. no. 105608, doi: [10.1016/j.bspc.2023.105608](https://doi.org/10.1016/j.bspc.2023.105608).
- [35] S. P. Kumar, J. Selvaraj, and A. Sahayadhas, "A comparative study on different measures to identify driver states," in *Proc. IEEE Int. Conf. Syst., Comput., Autom. Netw. (ICSCAN)*, Mar. 2019, pp. 1–5, doi: [10.1109/ICSCAN.2019.8878699](https://doi.org/10.1109/ICSCAN.2019.8878699).
- [36] J. A. Healey and R. W. Picard, "Detecting stress during real-world driving tasks using physiological sensors," *IEEE Trans. Intell. Transp. Syst.*, vol. 6, no. 2, pp. 156–166, Jun. 2005, doi: [10.1109/TITS.2005.848368](https://doi.org/10.1109/TITS.2005.848368).
- [37] H. U. Rehman, M. Naeem, M. Khan, G. Sikander, and S. Anwar, "Eye tracking based real-time non-interfering driver fatigue detection system," in *Proc. 10th Int. Conf. Electron., Comput. Artif. Intell. (ECAI)*, Jun. 2018, pp. 1–5, doi: [10.1109/ECAI.2018.8678951](https://doi.org/10.1109/ECAI.2018.8678951).
- [38] T. Liu, Y. Yang, G.-B. Huang, Y. K. Yeo, and Z. Lin, "Driver distraction detection using semi-supervised machine learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 4, pp. 1108–1120, Apr. 2016, doi: [10.1109/TITS.2015.2496157](https://doi.org/10.1109/TITS.2015.2496157).
- [39] W. Zhou, Z. Jia, C. Feng, H. Lu, F. Lyu, and L. Li, "Towards driver distraction detection: A privacy-preserving federated learning approach," *Peer-Rev. Netw. Appl.*, vol. 17, no. 2, pp. 896–910, Mar. 2024, doi: [10.1007/s12083-024-01639-5](https://doi.org/10.1007/s12083-024-01639-5).
- [40] Y. Chu, Z. Yuan, and P. L. Liao, "Privacy concern and acceptability of driver monitoring system," *Emerg. Technol. Future Work, AHFE*, vol. 117, pp. 195–205, Sep. 2023, doi: [10.54941/ahfe1004419](https://doi.org/10.54941/ahfe1004419).
- [41] B. Goel, A. K. Dey, P. Bharti, K. B. Ahmed, and S. Chellappan, "Detecting distracted driving using a wrist-worn wearable," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. Workshops (PerCom Workshops)*, Mar. 2018, pp. 233–238, doi: [10.1109/PERCOMW.2018.8480282](https://doi.org/10.1109/PERCOMW.2018.8480282).
- [42] B.-L. Lee, B.-G. Lee, and W.-Y. Chung, "Standalone wearable driver drowsiness detection system in a smartwatch," *IEEE Sensors J.*, vol. 16, no. 13, pp. 5444–5451, Jul. 2016, doi: [10.1109/JSEN.2016.2566667](https://doi.org/10.1109/JSEN.2016.2566667).

- [43] B. Li, J. Chen, Z. Huang, H. Wang, J. Lv, J. Xi, J. Zhang, and Z. Wu, "A new unsupervised deep learning algorithm for fine-grained detection of driver distraction," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 19272–19284, Oct. 2022, doi: [10.1109/TITS.2022.3166275](https://doi.org/10.1109/TITS.2022.3166275).
- [44] T. Huang and R. Fu, "Driver distraction detection based on the true driver's focus of attention," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 19374–19386, Oct. 2022, doi: [10.1109/TITS.2022.3166208](https://doi.org/10.1109/TITS.2022.3166208).
- [45] N. N. Pandey and N. Babu Muppalaneni, "Driver distraction identification using multiple machine learning approaches," in *Proc. Int. Conf. Comput. Intell. Comput. Appl. (ICCICA)*, Nov. 2021, pp. 1–6, doi: [10.1109/ICCICA52458.2021.9697193](https://doi.org/10.1109/ICCICA52458.2021.9697193).
- [46] A. Ezzouhri, Z. Charouh, M. Ghogho, and Z. Guennoun, "Robust deep learning-based driver distraction detection and classification," *IEEE Access*, vol. 9, pp. 168080–168092, 2021, doi: [10.1109/ACCESS.2021.3133797](https://doi.org/10.1109/ACCESS.2021.3133797).
- [47] A. Gumaci, M. Al-Rakhani, M. M. Hassan, A. Alamri, M. Alhussein, M. A. Razzaque, and G. Fortino, "A deep learning-based driver distraction identification framework over edge cloud," *Neural Comput. Appl.*, vol. 2020, pp. 1–16, Sep. 2020, doi: [10.1007/s00521-020-05328-1](https://doi.org/10.1007/s00521-020-05328-1).
- [48] G. K. Sahoo, S. K. Das, and P. Singh, "A deep learning-based distracted driving detection solution implemented on embedded system," *Multimedia Tools Appl.*, vol. 82, no. 8, pp. 11697–11720, Mar. 2023, doi: [10.1007/s11042-022-13450-6](https://doi.org/10.1007/s11042-022-13450-6).
- [49] F. Sajid, A. R. Javed, A. Basharat, N. Kryvinska, A. Afzal, and M. Rizwan, "An efficient deep learning framework for distracted driver detection," *IEEE Access*, vol. 9, pp. 169270–169280, 2021, doi: [10.1109/ACCESS.2021.3138137](https://doi.org/10.1109/ACCESS.2021.3138137).
- [50] M. Jogin, Mohana, M. S. Madhulika, G. D. Divya, R. K. Meghana, and S. Apoorva, "Feature extraction using convolution neural networks (CNN) and deep learning," in *Proc. 3rd IEEE Int. Conf. Recent Trends Electron., Inf. Commun. Technol. (RTEICT)*, May 2018, pp. 2319–2323, doi: [10.1109/RTEICT42901.2018.9012507](https://doi.org/10.1109/RTEICT42901.2018.9012507).
- [51] K. Nakano and B. Chakraborty, "Real-time distraction detection from driving data based personal driving model using deep learning," *Int. J. Intell. Transp. Syst. Res.*, vol. 20, no. 1, pp. 238–251, Apr. 2022, doi: [10.1007/s13177-021-00288-9](https://doi.org/10.1007/s13177-021-00288-9).
- [52] R. Sayed and A. Eskandarian, "Unobtrusive drowsiness detection by neural network learning of driver steering," *Proc. Inst. Mech. Eng., D, J. Automobile Eng.*, vol. 215, no. 9, pp. 969–975, Sep. 2001, doi: [10.1243/0954407011528536](https://doi.org/10.1243/0954407011528536).
- [53] F. Friedrichs and B. Yang, "Drowsiness monitoring by steering and lane data based features under real driving conditions," in *Proc. 18th Eur. Signal Process. Conf.*, Aug. 2010, pp. 209–213. Accessed: Aug. 05, 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/7096521>
- [54] Z. Li, S. Li, R. Li, B. Cheng, and J. Shi, "Online detection of driver fatigue using steering wheel angles for real driving conditions," *Sensors*, vol. 17, no. 3, p. 495, Mar. 2017, doi: [10.3390/s17030495](https://doi.org/10.3390/s17030495).
- [55] M. Patel, S. K. L. Lal, D. Kavanagh, and P. Rossiter, "Applying neural network analysis on heart rate variability data to assess driver fatigue," *Expert Syst. Appl.*, vol. 38, no. 6, pp. 7235–7242, Jun. 2011, doi: [10.1016/j.eswa.2010.12.028](https://doi.org/10.1016/j.eswa.2010.12.028).
- [56] M. V. M. Yeo, X. Li, K. Shen, and E. P. V. Wilder-Smith, "Can SVM be used for automatic EEG detection of drowsiness during car driving?" *Saf. Sci.*, vol. 47, no. 1, pp. 115–124, Jan. 2009, doi: [10.1016/j.ssci.2008.01.007](https://doi.org/10.1016/j.ssci.2008.01.007).
- [57] M. F. F. M. Hanafi, M. S. F. M. Nasir, S. Wani, R. A. A. Abdulghafor, Y. Gulzar, and Y. Hamid, "A real time deep learning based driver monitoring system," *Int. J. Perceptive Cognit. Comput.*, vol. 7, no. 1, pp. 79–84, Jul. 2021.
- [58] M. Zahid Hasan, J. Chen, J. Wang, M. S. Rahman, A. Joshi, S. Velipasalar, C. Hegde, A. Sharma, and S. Sarkar, "Vision-language models can identify distracted driver behavior from naturalistic videos," 2023, *arXiv:2306.10159*.
- [59] Z. Zhuang, Y.-K. Wang, Y.-C. Chang, J. Liu, and C.-T. Lin, "A connectivity-aware graph neural network for real-time drowsiness classification," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 32, pp. 83–93, 2024, doi: [10.1109/TNSRE.2023.3336897](https://doi.org/10.1109/TNSRE.2023.3336897).
- [60] F. Faul, E. Erdfelder, A. Buchner, and A.-G. Lang, "Statistical power analyses using G power 3.1: Tests for correlation and regression analyses," *Behav. Res. Methods*, vol. 41, no. 4, pp. 1149–1160, Nov. 2009, doi: [10.3758/brm.41.4.1149](https://doi.org/10.3758/brm.41.4.1149).
- [61] N. von Janczewski, J. Wittmann, A. Engeln, M. Baumann, and L. Krauß, "A meta-analysis of the n-back task while driving and its effects on cognitive workload," *Transp. Res. F, Traffic Psychol. Behav.*, vol. 76, pp. 269–285, Jan. 2021, doi: [10.1016/j.trf.2020.11.014](https://doi.org/10.1016/j.trf.2020.11.014).
- [62] V. Jeyhani, M. Mantysalo, K. Noponen, T. Seppanen, and A. Vehkaoja, "Effect of different ECG leads on estimated R-R intervals and heart rate variability parameters," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Jul. 2019, pp. 3786–3790, doi: [10.1109/EMBC.2019.8857954](https://doi.org/10.1109/EMBC.2019.8857954).
- [63] J. D. Van Der Laan, A. Heino, and D. De Waard, "A simple procedure for the assessment of acceptance of advanced transport telematics," *Transp. Res. C, Emerg. Technol.*, vol. 5, no. 1, pp. 1–10, Feb. 1997. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0968090X96000253>
- [64] R. Agarwal and J. Prasad, "A conceptual and operational definition of personal innovativeness in the domain of information technology," *Inf. Syst. Res.*, vol. 9, no. 2, pp. 204–215, Jun. 1998, doi: [10.1287/isre.9.2.204](https://doi.org/10.1287/isre.9.2.204).
- [65] S. V. Wass, T. J. Smith, and M. H. Johnson, "Parsing eye-tracking data of variable quality to provide accurate fixation duration estimates in infants and adults," *Behav. Res. Methods*, vol. 45, no. 1, pp. 229–250, Mar. 2013, doi: [10.3758/s13428-012-0245-6](https://doi.org/10.3758/s13428-012-0245-6).
- [66] D. A. Huffman, "The synthesis of sequential switching circuits," *J. Franklin Inst.*, vol. 257, no. 3, pp. 161–190, Mar. 1954.
- [67] H. Lee, R. Grosse, R. Ranganath, and A. Y. Ng, "Convolutional deep belief networks for scalable unsupervised learning of hierarchical representations," in *Proc. 26th Annu. Int. Conf. Mach. Learn.* New York, NY, USA: ACM, Jun. 2009, pp. 609–616, doi: [10.1145/1553374.1553453](https://doi.org/10.1145/1553374.1553453).
- [68] S. Kiranyaz, O. Avci, O. Abdeljaber, T. Ince, M. Gabbouj, and D. J. Inman, "1D convolutional neural networks and applications: A survey," *Mech. Syst. Signal Process.*, vol. 151, Apr. 2021, Art. no. 107398, doi: [10.1016/j.ymssp.2020.107398](https://doi.org/10.1016/j.ymssp.2020.107398).
- [69] A. Dempster, D. F. Schmidt, and G. I. Webb, "MiniRocket: A very fast (Almost) deterministic transform for time series classification," in *Proc. 27th ACM SIGKDD Conf. Knowl. Discovery Data Mining*. New York, NY, USA: Association for Computing Machinery, Aug. 2021, pp. 248–257, doi: [10.1145/3447548.3467231](https://doi.org/10.1145/3447548.3467231).
- [70] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.
- [71] A. van den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, and K. Kavukcuoglu, "WaveNet: A generative model for raw audio," 2016, *arXiv:1609.03499*.
- [72] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.* Red Hook, NY, USA: Curran Associates, 2017, pp. 1–12. Accessed: Aug. 5, 2024. [Online]. Available: <https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html>
- [73] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778, doi: [10.1109/CVPR.2016.90](https://doi.org/10.1109/CVPR.2016.90).
- [74] *Hands-On Machine Learning With Scikit-Learn, Keras, and TensorFlow*. Accessed: Jan. 2, 2025. [Online]. Available: <https://www.oreilly.com/library/view/hands-on-machine-learning/9781492032632/>
- [75] K. Benidis, S. S. Rangapuram, V. Flunkert, Y. Wang, D. Maddix, C. Turkmen, J. Gasthaus, M. Bohlke-Schneider, D. Salinas, L. Stella, F.-X. Aubet, L. Callot, and T. Januschowski, "Deep learning for time series forecasting: Tutorial and literature survey," *ACM Comput. Surv.*, vol. 55, no. 6, pp. 121:1–121:36, Dec. 2022, doi: [10.1145/3533382](https://doi.org/10.1145/3533382).
- [76] A. Dabbous, R. Berta, M. Fresta, H. Ballout, L. Lazzaroni, and F. Bellotti, "Bringing intelligence to the edge for structural health monitoring: The case study of the Z24 bridge," *IEEE Open J. Ind. Electron. Soc.*, vol. 5, pp. 781–794, 2024, doi: [10.1109/OJIES.2024.3434341](https://doi.org/10.1109/OJIES.2024.3434341).
- [77] P. Domingos, "A few useful things to know about machine learning," *Commun. ACM*, vol. 55, no. 10, pp. 78–87, Oct. 2012, doi: [10.1145/2347736.2347755](https://doi.org/10.1145/2347736.2347755).
- [78] D. J. J. Braithwaite, "A guide for analysing electrodermal activity (EDA) & skin conductance responses (SCRs) for psychological experiments," *Psychophysiology*, vol. 49, no. 1, pp. 1017–1034, 2013.
- [79] W. Morales-Alvarez, O. Sipele, R. Léberon, H. H. Tadjine, and C. Olaverri-Monreal, "Automated driving: A literature review of the take over request in conditional automation," *Electronics*, vol. 9, no. 12, p. 2087, Dec. 2020, doi: [10.3390/electronics9122087](https://doi.org/10.3390/electronics9122087).
- [80] J. Son, M. Park, and B. B. Park, "The effect of age, gender and roadway environment on the acceptance and effectiveness of advanced driver assistance systems," *Transp. Res. F, Traffic Psychol. Behav.*, vol. 31, pp. 12–24, May 2015, doi: [10.1016/j.trf.2015.03.009](https://doi.org/10.1016/j.trf.2015.03.009).

- [81] S. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," 2017, *arXiv:1705.07874*.
- [82] O. Loyola-González, "Black-box vs. white-box: Understanding their advantages and weaknesses from a practical point of view," *IEEE Access*, vol. 7, pp. 154096–154113, 2019, doi: [10.1109/ACCESS.2019.2949286](https://doi.org/10.1109/ACCESS.2019.2949286).
- [83] M. Marcano, F. Tango, J. Sarabia, A. Castellano, J. Pérez, E. Irigoyen, and S. Díaz, "From the concept of being 'the Boss' to the idea of being 'a team': The adaptive co-pilot as the enabler for a new cooperative framework," *Appl. Sci.*, vol. 11, no. 15, p. 6950, Jul. 2021, doi: [10.3390/app11156950](https://doi.org/10.3390/app11156950).
- [84] M. Marcano, S. Díaz, J. Pérez, A. Castellano, E. Landini, F. Tango, and P. Burgio, "Human-automation interaction through shared and traded control applications," in *Intelligent Human Systems Integration*. Cham, Switzerland: Springer, 2020, pp. 653–659, doi: [10.1007/978-3-030-39512-4_101](https://doi.org/10.1007/978-3-030-39512-4_101).



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